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Lagged Determinants of U.S. Output Growth

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Abstract

This paper presents two interactions based on Okun's Law and the Phillips Curve as potential determinants of the U.S. output growth. Alongside other macroeconomic indicators, these interactions are included in an OLS-estimated linear regression. After reaching a satisfactory model specification, several hypotheses regarding the significance and signs of the interaction coefficients are confirmed. As a result, we observe a significant and negative coefficient for the Phillips interaction, while the Okun interaction is found to be insignificant as per our hypothesis.

Contents

1	Introduction	3
1.1	Overview	3
1.2	Hypothesis	3
1.3	Theoretical Background	4
1.4	Literature Review	5
2	Data	6
3	Methodology	9
3.1	Data Preparation and Examination	9
3.1.1	Design Matrix Formation	9
3.1.2	Regressor Examination	10
3.2	BLUE Evaluation	10
3.3	Post-Fit Validation	11
3.3.1	Coefficient Significance	11
3.3.2	Residual Analysis	11
4	Results	12
4.1	Data Examination	12
4.2	Model Estimation	13
4.3	Pre-COVID Re-Estimation	15
4.4	Shock Adjusted Model Re-Estimation	17
4.5	Residual Diagnostics	19
4.5.1	Normality	19
4.5.2	Homoskedasticity	20
4.5.3	Autocorrelation	20
4.5.4	Stationarity	21
4.6	Gauss–Markov Conditions	21
4.7	Coefficient Significance and Interpretation	21
4.7.1	Interpretation of Coefficients	21
4.7.2	Joint Significance of Coefficients	23
5	Conclusion	23
6	Bibliography	24
7	Remarks	25
7.1	On Variable Selection	25

1 Introduction

1.1 Overview

Real GDP is a prominent measure of output due to its simplicity, interpretability, and effectiveness in communicating the overall activity of an economy. Alongside academic settings, Real GDP has retained its relevance in politics as one of the go-to measures for public communications regarding economic growth and prosperity. As per its wide-spread adoption, the measure has accumulated a plethora of literature referencing it from a multitude of perspectives.

This paper explores the link between GDP growth and theoretically grounded interactions of GDP deficit, unemployment gap, and output gap, alongside other exogenous regressors. Ultimately, the paper aims to show that empirical observations of Real GDP growth in the U.S. economy can be modelled optimally with a **Gauss–Markov** compliant **Distributed Lag Model** (DLM) specification where the exogenous contributors are theoretical descriptors of output.

1.2 Hypothesis

In the history of economic equilibria, multiple models have proposed mathematically defined relationships between Real GDP and many indicators. Moreover, some interactions between said indicators have been formalized into economic models. This paper focuses on testing the predictive and explanatory power of two such indicators involving unemployment gap, output gap, and GDP deflator (P_t^{GDP}):

- Unemployment Gap and Output Gap (**Okun-like**): $O_t = (u_t - u_t^*) \times (Y_t - Y_t^*)$
- Unemployment Gap and GDP Deflator (**Phillips-like**): $\Pi_t = (u_t - u_t^*) \times \Delta \log(P_t^{GDP})$

Alongside other exogenous regressors, this paper hypothesizes that a model of the form:

$$\Delta \log(Y_t) = C + \sum_{i=1}^n \beta_i X_{[i, t-1]} + \varepsilon_t \quad (1)$$

where $X_{[i, t-1]}$ includes the above defined interactions has an OLS estimator with no strong evidence against **BLUE**.

By Okun’s Law, we expect the output gap to be modelled by the unemployment gap and a negative coefficient. Therefore, the interaction O_t should be a redundant variable in the presence of any main-effects. On the other hand, the Π_t interaction is the only variable that brings an inflationary aspect into the model equation. Given the added information, we expect Π_t to add explanatory power though inflation while being supported by the in-set main effect of the unemployment gap. With these theorized expectations, build upon the design hypothesis and further hypothesize that O_t

should have an insignificant, negative (if non-zero) coefficient while Π_t should have a significant and negative coefficient.

1.3 Theoretical Background

We consider two equations to base our selection of interactions. First, the **Phillips Curve** describes the relationship between inflation and unemployment. It can be expressed in terms of expected inflation and the unemployment gap as follows:

$$\pi_t = \mathbb{E}_t[\pi_{t+1}] - \alpha(u_t - u_t^*) + \varepsilon_t \quad (2)$$

where π_t is the inflation rate at time t , $\mathbb{E}_t[\pi_{t+1}]$ is the expected inflation rate at time $t + 1$, u_t is the unemployment gap, α is a positive constant, and ε_t is a random error term. Equation 2 displays a negative relationship between inflation and unemployment gap with a factor of α . The inflation rate π_t in logarithmic space can be approximated by $\log(1 + \pi_t) = \Delta \log(P_t)$ for sufficiently small π_t . (up to 8-10%) The presented interaction substitutes the deflator for price levels and uses $\Delta \log(P_t^{GDP})$ as a deflator-based inflation term. In principle, we expect (and hypothesize) that the pseudo-inflation term defined will again have a negative relationship with unemployment gap with a non-zero factor.

We can use the Phillips Curve to connect inflation and unemployment; however, the connection of this interaction to our regression target is not yet established. Utilizing the **Fischer Equation**, we can relate inflation to nominal interest rates, and express output as a function of inflation using the **IS Curve**.

$$\begin{aligned} r_t &= i_t - \pi_t \\ \Rightarrow Y_t &= Y^* - \alpha(r_t - r_t^*) \end{aligned} \quad (3)$$

Having established Y_t as a function of π_t , we can expect to see a relationship in growth rates as well.

Next, we consider **Okun's Law**, which describes the relationship between unemployment and output (Real GDP):

$$\frac{(Y_t - Y_t^*)}{Y_t^*} = -\gamma(u_t - u_t^*) \quad (4)$$

where Y_t is the actual output, Y_t^* is the potential output, γ is a positive constant, and $(u_t - u_t^*)$ is the unemployment gap. Okun's Law can be algebraically rearranged to express Y_t in levels:

$$\begin{aligned} \frac{Y_t}{Y_t^*} - 1 &= -\gamma(u_t - u_t^*) \\ 100 \times Y_t &= Y_t^*(1 - \gamma(u_t - u_t^*)) \end{aligned} \quad (5)$$

The above transformation connects output to unemployment and establishes a negative relationship. A simple yet fundamental observation is that non-zero relationships of P_t , and $(u_t - u_t^*)$ are clearly defined in theory. However, we also observe that given direct relationship between unemployment gap and output gap is linearly defined without the need of the specific interaction O_t . In terms of gap modelling, we could conclude that these observations makes O_t directly redundant. Yet, in a growth rate context, we believe there's benefit in testing how the Okun based dynamics are mapped outside the gap framework.(Ferrentino and Vota, 2020) Interactions defined with these variables are therefore potential candidates to explain the output growth. (Although we expect one to be of no significance.) In any case, an empirical relationship can be analyzed to determine the differences between theoretical and observed interactions.

1.4 Literature Review

GDP growth is a topic of constant interest and numerous studies have explored its determinants in linear, non-linear, and machine learning contexts. (Maccarrone et al., 2021; Mandel and Liebens, 2019) With the two main drivers of this paper's hypotheses being the classical Philips Curve and Okun's Law; our take on GDP modelling can be called overly-conservative or outdated by modern standards. There are interesting, modern takes on both of the relationships these drivers describe. Although our focus is on re-visiting the long-established theories, iterations on the underlying ideas continue to prove their relevance.

For instance, "A Growth-Augmented Phillips Curve" (Tauber and Van Zandweghe, 2020) adds a growth term with its own coefficient to the traditional New Keynesian Phillips Curve (NKPC). The paper then runs historical parameter estimations to jointly estimate the coefficients of unemployment gap and growth using OLS. Published in 2020Q2, the paper does not include the COVID-19 pandemic in the estimation period. However, the shocks are well-within their out-of-sample forecast horizon. With their last observation being one period before the main COVID-19 shock, the paper manages to forecast the initial deflation and subsequent bounce-back relatively well. They predict a -35% change in growth and a +12% in the first response quarter. Papers such as these show that even purely theoretical, single-equation, and linear models can still hold noteworthy explanatory and predictive power in modern settings.

Another example of a modern approach is presented by Ferrentino and Vota, 2020 where the authors propose a static mathematically derived relationship between both the Philips Curve - Output Growth and Okun's Law - Output Growth. The paper connects the two relationships to net export dynamics via the output growth equations. Before any estimation, they propose a NX growth estimator consisting of multiple cyclical functions including the sin and cos of output growth as regressors, instead of the standard business cycle approach of using the output gap.

While the above papers present modern twists on the concepts this paper builds upon, they also show that Okun’s Law and the Philips Curve are still perfectly valid starting points for research. Both papers utilize OLS estimation in single-equation scenarios, starting from the most basic forms of the relationships. The interactions and hypotheses proposed in this paper are akin to a simplified version of the above methodologies, where the focus is on testing the validity of interactions that are present in theory as empirical predictors.

2 Data

The initial model specification includes a wide range of macroeconomic indicators alongside the interactions. Observations for all series span from 1989Q4 to 2025Q1 and the frequency is quarterly. All series are sourced from the St. Louis Federal Reserve Economic Data (FRED) database.

Variable	Units	FRED ID
Real GDP (Y)	Billions of Chained 2017 Dollars (Adjusted)	GDPC1
Personal Consumption Expenditures (pce)	Billions of Chained 2017 Dollars (Adjusted)	PCECC96
Private Domestic Investment (pdi)	Billions of Chained 2017 Dollars (Adjusted)	GPDIC1
Government Expenditures ($govexp$)	Billions of Chained 2017 Dollars (Adjusted)	GCEC1
Output Deflator ($odef$)	Chained Index 2017=100 (Adjusted)	GDPDEF
Natural Unemployment Rate (u^*)	% (Not Adjusted)	NROU
Unemployment Rate (u)	% (Adjusted)	LRUN64TTUSQ156S
Potential Output (Y^*)	Billions of Chained 2017 Dollars (Adjusted)	GDPPOT

Table 1: Macroeconomic series used in the model along with their FRED IDs.

Using the series in Table 1, we construct our target variable and regressors as follows:

- Target Variable: $\Delta \log(Y_t) \times 100$
- Regressors:

$$X_{[t, 1]} (pce): \Delta \log(pce_{t-1})$$

$$X_{[t, 2]} (pdi): \Delta \log(pdi_{t-1})$$

$$X_{[t, 3]} (govexp): \Delta \log(govexp_{t-1})$$

$$X_{[t, 4]} (ugap): \Delta(u_{t-1} - u_{t-1}^*)$$

$$X_{[t, 5]} (ogap): \log(Y_{t-1}) - \log(Y_{t-1}^*)$$

$$X_{[t, 6]} (okun): (u_{t-1} - u_{t-1}^*) \times ogap$$

$$X_{[t, 7]} (phillips): (u_{t-1} - u_{t-1}^*) \times \Delta \log(odef_{t-1})$$

All variables that are observed in levels are transformed into log differences while both the gap variables and interactions are not differenced. (except for *ugap* due to multicollinearity concerns¹.) In our specification, log differencing is used to clean the regressors of slow-moving trends and potential multicollinearity; which is mostly an artifact of data in pure levels. The gaps and interactions are already detrended to an extent by their construction.

¹See Remark 7.1 for the implications and reasoning.

Distributions of Regressors

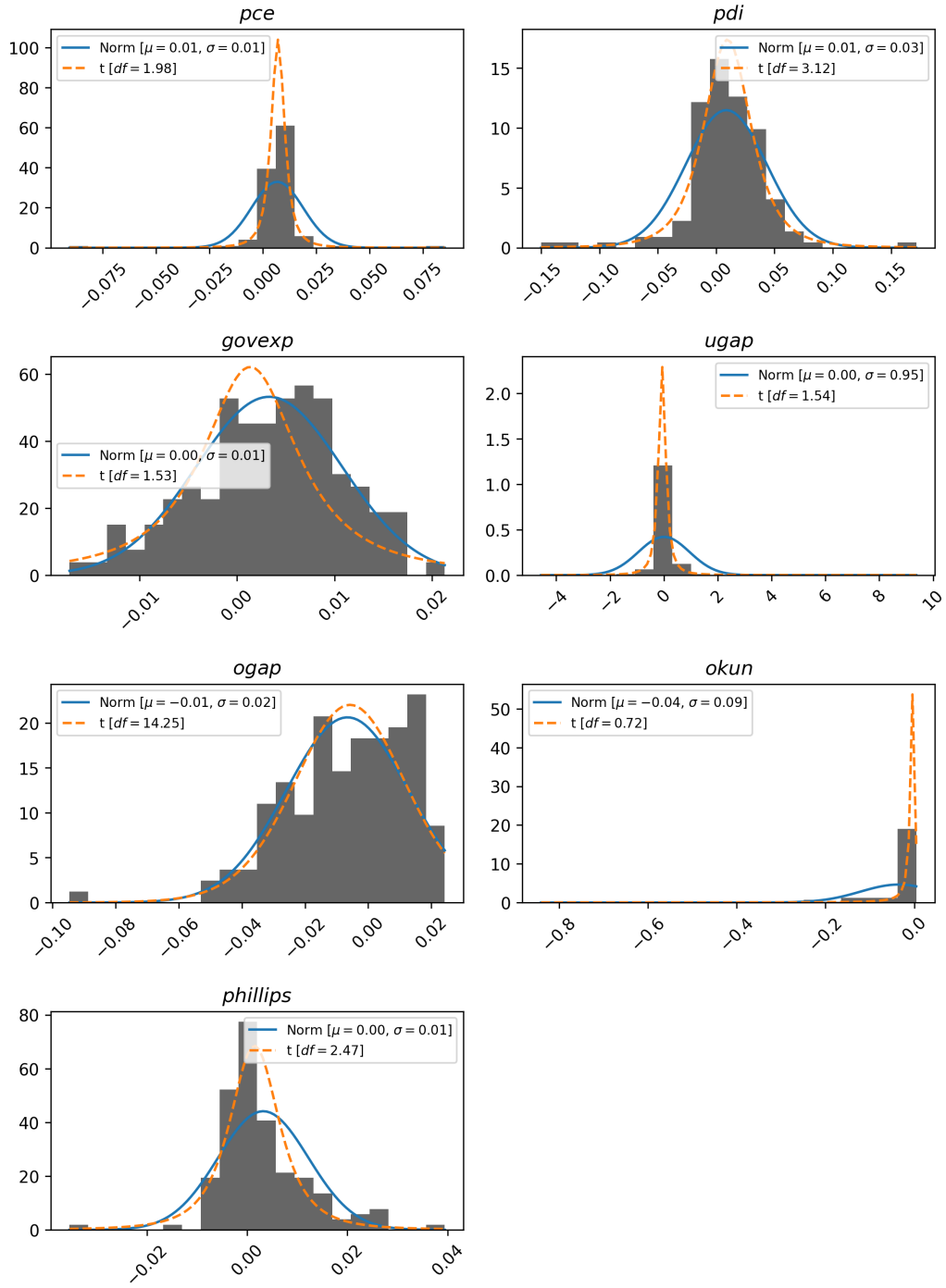


Figure 1: Density histograms of regressors.

3 Methodology

The proposed methodology of this paper can be split into four main components:

- Data Preparation and Examination
- Model Estimation
- BLUE Evaluation
- Post-Fit Validation

Each component is detailed in the following subsections.

3.1 Data Preparation and Examination

The preparation step consists of using the series defined in Table 1 to create regressors and target as listed. Vectorized transformations are applied to the raw data to create the necessary variables. The differencing process requires each observation to have a predecessor; therefore the first observation is lost in the transformation. This pushes our observation window from 1989Q4–2025Q1 to 1990Q1–2025Q1.

After the variables are constructed, we shift all regressors by one period to produce the final dataset. This ensures that all regressors at time t are known; making the model have predictive validity rather than being a purely explanatory specification. Having shifted the regressors, we again lose the first observation in the dataset, pulling the final observation window to 1990Q2–2025Q1 with a total of 140 observations.

3.1.1 Design Matrix Formation

X is formed by converting the model equation specified in Equation 1 into matrix form. The inner product operation of OLS estimation can be written as a matrix multiplication of observations and coefficients:

$$\underbrace{\begin{bmatrix} 1 & X_{1,2} & \cdots & X_{1,m+1} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n,2} & \cdots & X_{n,m+1} \end{bmatrix}}_{\text{Design Matrix}} \times \begin{bmatrix} C \\ \beta_1 \\ \vdots \\ \beta_{m+1} \end{bmatrix} \quad (6)$$

Where n is the number of observations and m is the number of features (including the intercept). Using our observation count $n = 140$ and number of regressors $m = 7$, the design matrix X takes the dimensions of 140×8 .

3.1.2 Regressor Examination

While the model order will not be adjusted based on the results, the Cross Correlation Function (CCF) of each regressor with the target variable will be examined to confirm the appropriateness of the lag structure. The **Box–Jenkins Prewhitening** procedure will be applied to expose a clearer signal of the underlying cross-correlations. Box–Jenkins Prewhitening transforms the exogenous variables into residuals of their AR representation, then uses the β coefficient to re-scale the dependent variable. (Hackhard and Romer, 2025) Formally, the $AR(1)$ variant of the process can be shown as:

$$\begin{aligned}\hat{X}_i &= (L^1 X_i)\beta_i \\ \hat{y}_i &= y\beta_i \\ \rho_{y,i} &= \text{corr}(\hat{y}_i, X_i - \hat{X}_i)\end{aligned}\tag{7}$$

Where L^1 is the lag-1 operator.

Additionally, the **Variance Inflation Factor** (VIF) of all regressors will be checked to ensure multicollinearity is not present. VIF is computed as:

$$VIF_{X_i} = \frac{1}{1 - R_{X_i}^2}\tag{8}$$

Where $R_{X_i}^2$ is the coefficient of determination of the regression of X_i on all other columns of X . The common threshold for multicollinearity, $VIF > 10$, will be used.

3.2 BLUE Evaluation

After estimating the model coefficients, we evaluate the Gauss–Markov conditions to confirm whether the resulting fit is BLUE. Previous stages of the methodology involve checks for the pre-fit Gauss–Markov conditions:

- **Linearity of Parameters:** by definition the model employed is a linear combination of regressors and coefficients. Therefore, this condition is satisfied at the stage of model selection.
- **Full Rank:** $\text{rank}(X) = k$ must hold for $X^\top X$ to be invertible. Therefore, this condition must hold for the model to be estimable via OLS.

The remaining Gauss–Markov conditions concern ε and are verified post-estimation:

- **Spherical Errors:** This condition holds if $\text{Var}(\varepsilon) = \sigma^2 I_n$; implies that the residuals are homoskedastic and uncorrelated. The two criteria can be tested separately using the **Breusch–Pagan Test** for homoskedasticity and the **Breusch–Godfrey Test** for autocorrelation.

- **Strict Exogeneity:** In textbook terms strict exogeneity requires $\mathbb{E}[\varepsilon_i | X] = 0$. However, this condition is not directly testable. It’s often assumed to hold given weak exogeneity holds, and the design matrix X shows no direct signs of endogenous interactions. A **RESET Test** will be employed to check for model misspecification and omitted nonlinearities. Moreover, the **Harvey–Collier Test** will be used to further assess model specification. Strong exogeneity will then be assumed, and therefore any finding of BLUE will be referred to as “approximate” under the modelling assumptions.

3.3 Post-Fit Validation

After estimating the model and evaluating the Gauss–Markov conditions, regression coefficients and residuals will be evaluated. Alongside understanding variable significances and outliers, this step will potentially highlight the sources of issues that may arise with the BLUE evaluation.

3.3.1 Coefficient Significance

Coefficients will be tested for significance both individually and jointly. Individual significance will be tested with *t*-tests, while model-wise joint significance will be tested with an *F*-test. Additionally, the joint significance of coefficient subsets will be tested using **Wald Tests**.

3.3.2 Residual Analysis

The residuals of a model are very important both in predictive terms, and as tools of analyzing the model fit. Therefore, multiple aspects of the residuals will be examined before deeming the model valid/efficient.

To begin with, the fitted values will be plotted against the observed target to display the directionality and scale of residuals in a visual manner. This will be complemented by a QQ plot and a **Shapiro–Wilk Test** to assess the normality of residuals. Normality is not a Gauss–Markov condition, but it is important for inference.

Finally, potential outliers will be examined using **Leverage** and **Cook’s Distance**. Given the nature of our target, we can immediately expect that the 2008 Financial Crisis and 2020 COVID-19 shocks will be visible in this portion of the analysis. If found to be detrimental to the model generalization, removal of outliers will be considered as required.

4 Results

4.1 Data Examination

With the design matrix constructed as per Section 3.1.1, we can compute the inter-regressor interactions with VIF and the regressor-target relations with CCF. Our initial VIF computations are as follows:

Regressor	VIF
<i>pce</i>	4.79
<i>pdi</i>	2.05
<i>govexp</i>	1.53
<i>ugap</i>	9.22
<i>ogap</i>	3.94
<i>okun</i>	4.62
<i>phillips</i>	3.22

Table 2: Variance Inflation Factor (VIF) of all regressors.

All VIF values are below the threshold, and we can conclude that multicollinearity is not a concern in our design matrix.

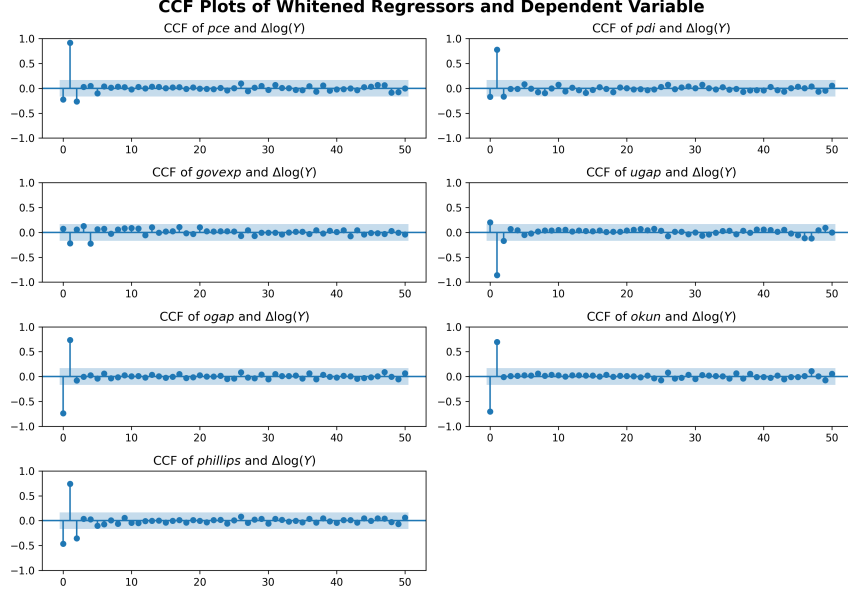


Figure 2: CCF plots of the whitened exogenous variables and the dependent variable.

The CCF results in Figure 4.1 show all regressors have significant lag-1 correlations, supporting the validity of the matrix specification with L^1 . As expected, *pce* and *pdi* show strong positive correlations while *ugap* shows a negative correlation as theorized. Moreover, almost all variables with significant lags have a sign changed significant correlation immediately after the initial lag that's significant. This might be an indicator of cyclical behavior, or “response functions” with overshooting. *ogap* and *okun* show particularly similar magnitudes and sign patterns. With *ogap* being a component of *okun*, we can hypothesize that *okun*'s correlation pattern is mostly inherited from *ogap* rather than *ugap*. (at least in the first 2 lags)

4.2 Model Estimation

Having confirmed the design matrix is sufficiently well-behaved, we proceed with the estimation of the initial model. The fit results in the coefficients:

Variable	Coefficient	t-test p	Significance
C	0.0288	0.8309	×
<i>pce</i>	76.8300	$\cong 0$	✓
<i>pdi</i>	9.7348	0.0229	✓
<i>govexp</i>	-22.6115	0.0730	×
<i>ugap</i>	1.1538	$\cong 0$	✓
<i>ogap</i>	-5.0182	0.5201	×
<i>okun</i>	-3.4977	0.0670	×
<i>phillips</i>	-32.7522	0.0315	✓
$\sum \checkmark = 4$			

Table 3: Estimated Coefficients of the Initial Model.

Table 3 shows a decent result with multiple coefficients being significant. We see that one of our interaction terms, *phillips*, has an individually significant coefficient while *okun* does not. The overall model fit statistics confirm this observation:

Statistic	Value
R^2	0.4071
Adjusted R^2	0.3703
RMSE	0.864
F-statistic	36.8205
p-value (F)	$\cong 0$
$X^\top \hat{\epsilon}$	$-8.49 \times 10^{-14} \cong 0$

Table 4: Overall model fit statistics.

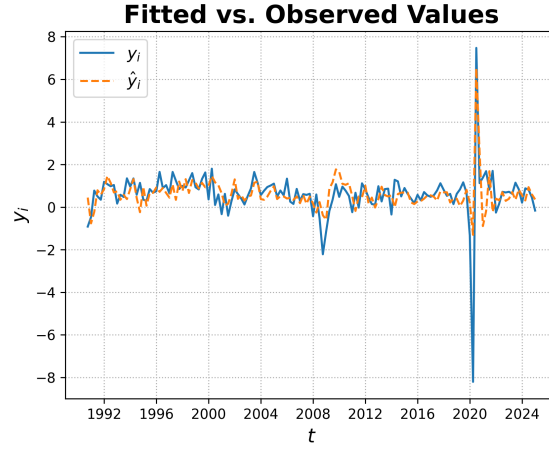


Figure 3: Fitted vs. Observed Values

With an R^2 of 40.71% and a passing F -test, the fit looks good at face value. However, the application of a RESET test tells a different story. The current model produces a p -value of 2.34×10^{-14} which is a strong indicator of misspecification. In fact, Figure 3 hints at a possible cause of this. With the 2008 Financial Crisis and 2020 COVID shocks in the data, we clearly see that the model goes out of its way to fit said points. This is especially visible in 2020Q3 where the model re-produces the extreme positive shock to output growth. This behavior is not expected from a generalized model and is potentially working against the model specification test. To confirm our suspicions of extreme outliers, we plot the Cook's Distance of all residuals:

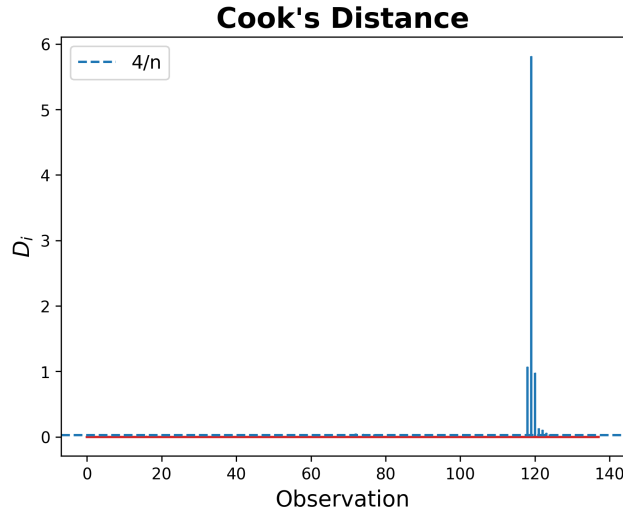


Figure 4: Cook's Distance of Initial Model Residuals.

We immediately see that the COVID-19 shocks dominate the influence statistics. $D_{2020Q3} = 5.81$ is orders of magnitude above the $4/n \approx 0.03$ threshold. In total, 10 observations exceed the threshold. 3 of these are attributable to the Financial Crisis (between 2008Q4 and 2010Q1) while the remaining 7 are consecutive observations from near-COVID (2020Q2–2021Q4).

In light of these results, it is clear that COVID-19 brings about undesirable fit-time behavior into the model. Of course, these observations are not to be discarded lightly, especially given time-continuity is crucial for preserving the interpretability of lagged models. However, given 7 consecutive observations with extreme influence on the model, there's a possibility of structural breaks. Therefore, two Chow tests were conducted with the break points at 2020Q2 and 2020Q3. Both tests confirmed the presence of structural breaks with $p_{2020Q2} = 0.01$ and $P_{2020Q3} \approx 0$. Seeing the results, we elected to re-estimate the current model using pre-COVID data before running further tests.

4.3 Pre-COVID Re-Estimation

After confirming the VIF of the pre-COVID design matrix cleared the thresholds ($VIF(X_i) < 10 \forall X_i \in X$), the model was re-estimated and the following coefficients were obtained:

Variable	Coefficient	t-test p	Significance
C	0.1934	0.0528	X
pce	63.5334	≈ 0	✓
pdi	4.5738	0.0190	✓
$govexp$	-10.7911	0.1622	X
$ugap$	-0.1312	0.5661	X
$ogap$	-13.8038	0.0043	✓
$okun$	-0.1075	0.9322	X
$phillips$	-32.7953	0.0015	✓
$\sum \checkmark = 4$			

Table 5: Estimated Coefficients of the Pre-COVID Model.

As in the previous estimation, Table 5 show four individually significant coefficients. Notably, $ogap$ becomes significant while $ugap$ loses its significance. The overall model fit statistics are as follows:

Statistic	Value
R^2	0.3955
Adjusted R^2	0.3507
RMSE	0.4525
F-statistic	8.83
p-value (F)	$\cong 0$
$X^\top \hat{\varepsilon}$	$-5.30 \times 10^{-14} \cong 0$

Table 6: Overall model fit statistics.

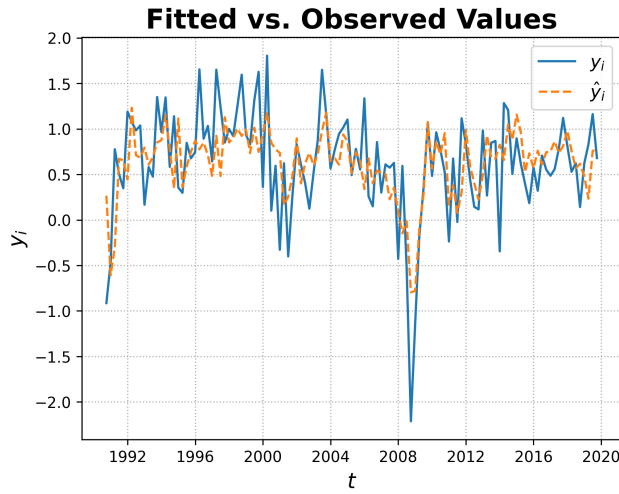


Figure 5: Fitted vs. Observed Values (Pre-COVID Model)

The fit statistics in Table 6 are comparable to the initial estimation, with a slight decrease in R^2 . However, Figure 5 again displays the regime-breaking 2008 shock. Knowing the model specification issues of the previous fit, we confirm D_{2008Q4} is well above the $4/n$ threshold before running the RESET test again. The test again results in a rejection with $p = 0.0167$. Although this is again a misspecification indicator, we observe that a sizable improvement in p -value has been achieved without explicitly adjusting any regressors. However, the strategy of pulling back the observations in this case would lead to a large loss of data. After evaluating the 2008Q4 point as a break point candidate in a Chow test, we found $p = 0.1427$ and could not confirm a break. Therefore, we proceed with the pre-COVID model while introducing a signal dummy $I_{2008Q4} = \mathbb{1}_{\{t=2008Q4\}}$. By this approach, we're effectively treating 2008Q4 as a single-period shock. The dummy provides the model with a single coefficient to deal with the 2008 shock exactly while having no effect on other observations. On the other hand, this choice introduces a loss in degrees of freedom (DoF) by increasing the X dimension.

4.4 Shock Adjusted Model Re-Estimation

For one more time, we confirm the VIF of all regressors (including the newly added dummy) are below threshold and re-estimate the model to see if the loss of DoF is justified. The estimation results in the coefficients:

Variable	Coefficient	t-test p	Significance
C	0.2714	0.0059	✓
pce	54.3276	$\cong 0$	✓
pdi	4.8823	0.0088	✓
$govexp$	-9.5442	0.1942	×
$ugap$	0.0241	0.9135	×
$ogap$	-10.5652	0.0234	✓
$okun$	0.2778	0.8179	×
$phillips$	-24.2331	0.0157	✓
I_{2008Q4}	-1.7381	0.0006	✓
$\sum \checkmark = 6$			

Table 7: Estimated Coefficients of the Pre-COVID Model.

Table 7 shows that the added dummy term is highly significant. Moreover, the addition has caused the intercept to become individually significant. The overall model fit statistics are as follows:

Statistic	Value
R^2	0.4577
Adjusted R^2	0.4121
RMSE	0.4286
F-statistic	10.0332
p-value (F)	$\cong 0$
$X^\top \hat{\varepsilon}$	$2.00 \times 10^{-14} \cong 0$

Table 8: Shock Adjusted model fit statistics.

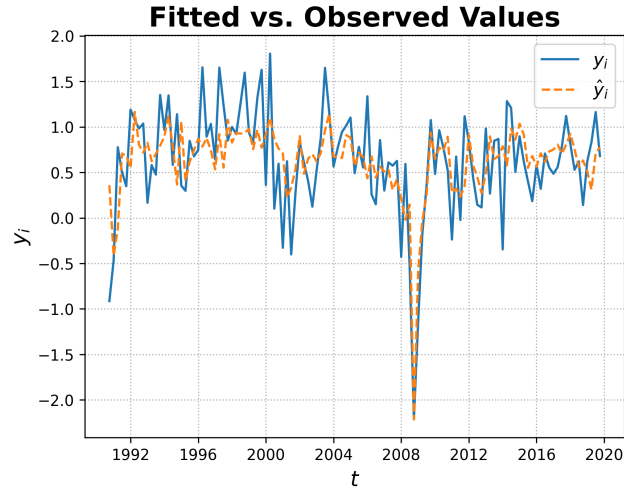


Figure 6: Fitted vs. Observed Values (Shock Adjusted Model)

The fit statistics in Table 8 show a sizable improvement in R^2 alongside an increase in the F -statistic. With the extravagant shock now accounted for, Figure 6 looks much better in terms of the presence of high-influence outliers. Plotting the Cook's Distance confirms this:

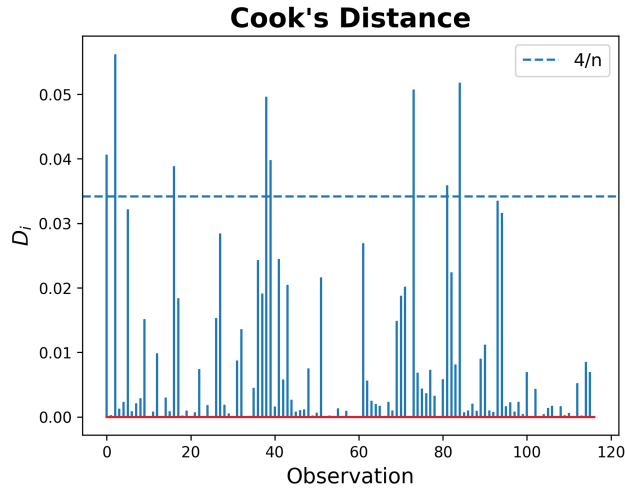


Figure 7: Cook's Distance of Shock Adjusted Model Residuals.

Although there still are 8 points above threshold, the distribution of D_i looks much more uniform and sensible. Running the RESET test one final time yields a p -value of 0.3837; indicating no evidence against correct model specification. With this confirmation, we proceed our analysis with the shock adjusted model.

4.5 Residual Diagnostics

Although the Cook's Distance plot displayed a basic look at the residuals, further diagnostics and tests are necessary to reach an accurate evaluation of the fit. Normality, Homoskedasticity, Autocorrelation, and Stationarity of residuals will be examined in this section.

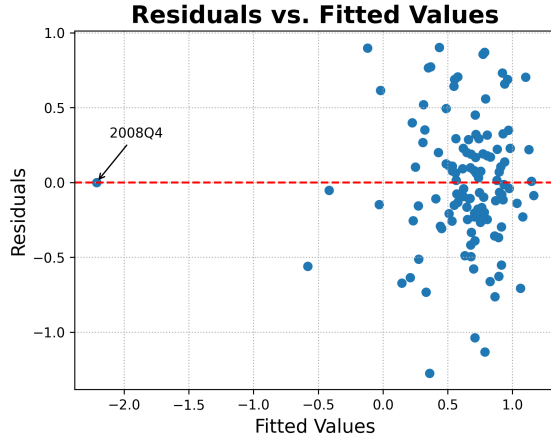


Figure 8: Residuals over Time.

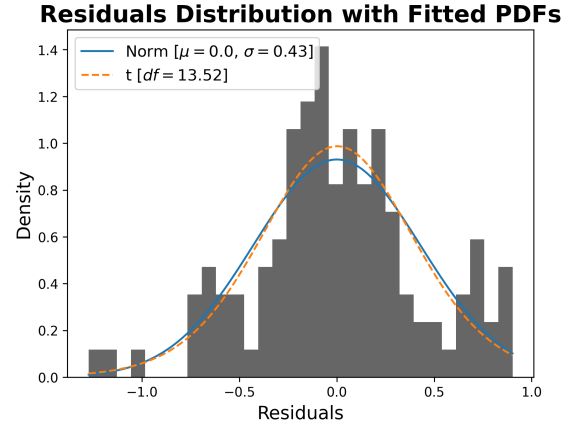


Figure 9: Histogram of Residuals.

Inspecting Figure 8, we see no extreme outliers except the 2008Q4. Which, as displayed by the 0 residual, is perfectly accounted for by the dummy. No emergent patterns are visible, and the residuals appear to be randomly scattered around 0.

Figure 9 shows a roughly normal distribution of residuals centered around 0. The parameter estimations suggest that $N(0, 0.43^2)$ and $t(13.52)$ are the best fitting distributions in their respective families.

4.5.1 Normality

Normality can both be tested for and examined visually. A QQ Plot alongside a Shapiro–Wilk test will be employed for this purpose.

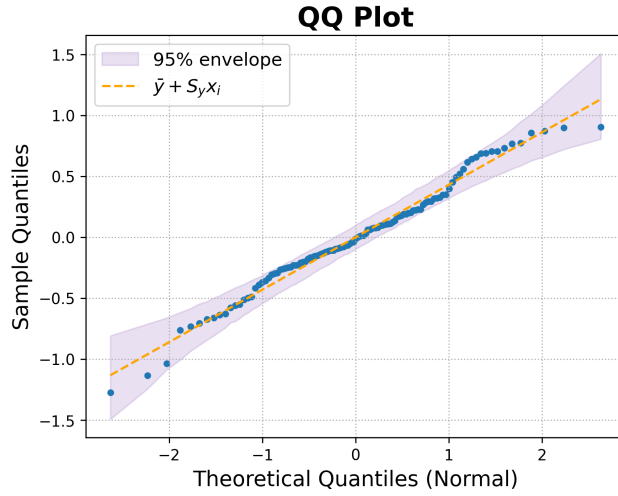


Figure 10: QQ Plot of Shock Adjusted Model Residuals.

The QQ Plot against normal quantiles shows that all residuals are contained within the 95% CI bands. This is a strong indicator of normality, which is further confirmed by the Shapiro–Wilk test results of $SW = 0.9825$ and $p = 0.1321$ failing to reject the null hypothesis of normality.

4.5.2 Homoskedasticity

Homoskedasticity is an important indicator of spherical errors and can be tested using the Breusch–Pagan test. A BP application on the shock adjusted model residuals yields $BP = 7.21$ and $p = 0.4853$; failing to reject the null hypothesis of homoskedasticity.

4.5.3 Autocorrelation

Autocorrelation of residuals indicates slow-moving patterns in the errors, which is undesirable and often results in inefficient estimates. To test for autocorrelation, we apply the Breusch–Godfrey test for lags $[1, 8]$ (2 years).

Lag	1	2	3	4	5	6	7	8
BG	5.00	5.57	6.93	8.57	8.67	9.84	10.19	10.77
p	0.975	0.938	0.926	0.927	0.877	0.868	0.822	0.785

Table 9: Breusch–Godfrey Test Results for Shock Adjusted Model Residuals.

All lags fail to reject the null hypothesis of no autocorrelation with strong p -values as seen in Table 9.

4.5.4 Stationarity

Finally, stationarity of residuals is an important aspect to consider in time series models. Non-stationary residuals can indicate omitted trends or cycles in the model specification. To test for stationarity, we apply the Augmented Dickey–Fuller (ADF) test on the residuals. The ADF test results in $ADF = -2.91$ and $p = 0.0036$; rejecting the null hypothesis of a unit root and indicating stationarity of residuals.

4.6 Gauss–Markov Conditions

Based on our results regarding the residual diagnostics and model specification tests, we can evaluate the approximate fulfillment of the GM conditions for the shock adjusted model:

- **Linearity:** The model is linear in parameters by design (Equation 1).
- **Full Rank:** We computed a lack of perfect multicollinearity with $VIF(X_i) < 5 \forall X_i \in X$ and $\text{cond}(X) = 75.64$. While a $\text{cond}(X) < 30$ is the commonly used diagnostic threshold, having multiple interactions in X increases the expected condition number by design. Given the low VIF values, we accept the fulfillment of the full rank condition.
- **Spherical Errors:** We found no evidence against homoskedasticity, no autocorrelation up to lag 8, and stationarity of residuals. Therefore, we accept the fulfillment of the spherical errors condition.
- **Strict Exogeneity:** While strict exogeneity is not testable, the model specification tests and residual diagnostics were all in favor of correct model specification. While said tests are definitively not sufficient for the strong claim of exogeneity, we believe that the evidence presented is supporting weak exogeneity. Although further work would be necessary to confirm this, we will not deny the possibility of BLUE on the basis of this uncertainty.

4.7 Coefficient Significance and Interpretation

With no strong proof against the GM conditions' fulfillment, we deem the OLS estimator of the shock adjusted model to be sufficiently close to **BLUE**. Therefore, we can analyze the individual coefficients, their joint significance, and confirm our hypotheses alongside an interpretation of the results.

4.7.1 Interpretation of Coefficients

Our model contains a mixed set of target-regressor interactions. With our target being in log growth space, Our coefficients will follow one of 3 interpretations:

- **$\Delta \log - \Delta \log$:** For regressors in log growth units, a percentage point (pp) increase in the regressor leads to an increase of β_i pp in the target’s growth rate. For example, if $\Delta \log(pce)$ increases from 0.02 to 0.03 (1 pp increase), then we expect to see a $1/100 \times \beta_{pce}$ pp increase in $\Delta \log(Y)$. (Our target is scaled by 100 to represent percentages, while the regressor is in fractional pp.)
- **$\Delta \log - \Delta gap$:** For regressors in gap difference units, +1 pp increase in the regressor leads to a $\beta_i \times 1/100$ pp increase in the target’s growth rate.
- **$\Delta \log - gap$:** For regressors in gap level units (including $\log(X_i) - \log(X_i^*)$), +1 unit increase in the regressor leads to a $+\beta_i$ pp increase in the target’s growth rate. However, when log points are used, the standard “1% increase” in gaps correspond to 0.01 unit increase. Therefore, we will use the more realistic increment of +0.01 for unit increase interpretations of this kind.²

Applying these interpretations to our variables, we have:

Variables			
Name	Type	Coefficient	Interpretation
C	Intercept	0.2714	+0.2714pp
pce	$\Delta \log - \Delta \log$	54.3276	$pce + 1pp \rightarrow 100\Delta \log(Y) + 0.5433pp$
pdi	$\Delta \log - \Delta \log$	4.8823	$pdi + 1pp \rightarrow 100\Delta \log(Y) + 0.0488pp$
$govexp$	$\Delta \log - \Delta \log$	-9.5442	$govexp + 1pp \rightarrow 100\Delta \log(Y) - 0.0954pp$
$ugap$	$\Delta \log - \Delta gap$	0.0241	$ugap + 1pp \rightarrow 100\Delta \log(Y) + 0.0241pp$
$ogap$	$\Delta \log - gap$	-10.5652	$ogap + 0.01 \rightarrow 100\Delta \log(Y) - 0.1057pp$
Interactions			
Name	Type ³	Coefficient	Interpretation
$okun$	$\Delta \log - gap$	0.2778	$okun + 0.01 \rightarrow 100\Delta \log(Y) + 0.002778pp$
$phillips$	$\Delta \log - gap$	-24.2331	$phillips + 0.01 \rightarrow 100\Delta \log(Y) - 0.2423pp$
I_{2008Q4}	Dummy	-1.7381	$t = 2008Q4 \rightarrow 100\Delta \log(Y) - 1.7381pp$

Table 10: Interpretation of Shock Adjusted Model Coefficients.

Table 10 summarizes the interpretations of all coefficients and displays 2 notable results. First, we observe that the pce coefficient is the largest contributor (excluding the dummy) to output growth within the regressors. This aligns with the expectations brought by the accounting identity of GDP. Moreover, we see that the derived interaction $phillips$ has a strong negative contribution to output, being the second most influential regressor in terms of percent influence.

²Refer to Figure 1 to inspect variable distributions and typical value ranges.

³Depending on the structure of the interactions, determining the type of interpretation dynamics can be difficult. Both $okun$ and $phillips$ are not necessarily of the type $\Delta \log - gap$, but are written as such because their interpretation aligns with the dynamics of a $\Delta \log - gap$ interaction.

Results of the interpretation align with the hypotheses outlined in Section 1.2. Both the significance and sign direction of *philips* and *okun* are as expected. However, an unexpected result is the insignificance of *ugap*. Seeing the *ogap* significance alongside *ugap* results, we can pose a complementary hypothesis that the output gap is a more direct determinant of output growth that captures the same dynamics *ugap* would.

4.7.2 Joint Significance of Coefficients

Looking at the model-wide F -test results in Table 8, we see that the shock adjusted model is jointly significant with $p \cong 0$. To further analyze the joint significance of variables, a series of regressor sub-sets were elected on the basis of theoretical adjacency.

- **Demand-Side Regressors:** $\{pce, pdi, goverp\}$
- **Gap/Cyclical Regressors:** $\{ugap, ogap\}$
- **Cross-Effect Regressors (Interactions):** $\{okun, phillips\}$

A Wald test was applied to each block to evaluate their joint significance in the model.

Regressor Block	Wald Statistic	p -value
Demand-Side Regressors	66.01	$\cong 0$
Gap/Cyclical Regressors	4.62	0.0119
Cross-Effect Regressors	2.27	0.1077

Table 11: Wald Test Results for Regressor Blocks.

Table 11 shows that both the demand-side and gap/cyclical regressor blocks are jointly significant. However, the cross-effect regressor block fails to reach joint significance at the 5% level. Given the individual significance of *philips*, we can hypothesize that *okun* is weakening the significance of the block. While unexpected, this result is not a direct contradiction to our hypotheses and the model's validity.

5 Conclusion

Although our initial design matrix indicated misspecification issues, we were able to address these through shock adjustments and the identification of a structural break. Ultimately, our final model included every regressor in the initial specification while clearing the diagnostic tests. Therefore, we believe our current specification is sufficient to clear our first hypothesis. (existence of an efficient model including the two interactions)

Our hypothesis regarding the insignificance of *okun* was confirmed. Evident by the lack of significance in both *okun* and *ugap*, we believe that with respect to output growth, the Okun-based influence is fully linear and captured by a single main effect (*ogap*). Further supporting this view is the sign, magnitude, and significance of the *ogap* coefficient. Being the third most influential regression coefficient by percent influence, *ogap* captures a decently-strong negative relationship, aligning with the underlying theory.

Our second hypothesis regarding the significance and sign of *philips* was also confirmed. Again, *philips* was one of the most influential regressors, being second only to *pce*. While the entirety of its influence cannot be attributed to the cross-effect alone⁴, it is clear that inflationary dynamics are important determinants of output growth.

Overall, we believe that our model was able to explain a decent portion of U.S. output growth dynamics for a prediction horizon of 1 quarter. Moreover, significant coefficients emerged both from the accounting identity (demand-side) and from the gaps. Although further robustness checks and extensions are necessary to solidify our claims, we believe that our findings provide useful insights into the lagged determinants of U.S. output growth.

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⁴See Remark 7.1

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7 Remarks

7.1 On Variable Selection

The design matrix intentionally omits two main effects that correspond to the interaction terms used in our hypotheses: the level unemployment gap ($u - u^*$) and inflation measured as ($\Delta \log(P^{GDP})$). We instead include $\Delta(u - u^*)$, incorporating $u - u^*$ and $\Delta \log(P^{GDP})$ only through interaction terms.

As a robustness check, we estimated an alternative specification that includes the two main effects. (On top of the shock adjusted model) The main-effect coefficients were found to be insignificant, and both interaction terms lost their significance as well. The coefficients of interest only showed marginal changes while the R^2 had a small increase from 0.45 to 0.47. However, these additions introduced severe multicollinearity issues, with $\text{cond}(X) = 698.2$ and $\max(VIF(X)) = 24.72$.

Therefore, we elected to use the specification with $\Delta \log(P^{GDP})$ omitted as a main-effect, and $u - u^*$ differenced in level form. We confirmed that the models were qualitatively very similar, while

the selected model had tangible benefits in terms of multicollinearity and the preservation of degrees of freedom. We acknowledge that excluding main effects makes interaction coefficients best viewed as reduced-form conditional relationships rather than clean structural cross-effects.